Emotion-Guided Cross-Attention for Multi-Task Abusive Language Detection

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***Abstract*—Intimate partner violence and psychologically unhealthy relationship dynamics often begin with subtle emotional abuse in everyday language, making early detection from text both critical and challenging. Existing abuse detection systems often focus on coarse-grained or binary labels and only partially capture nuanced behaviors such as hostility, condescension, or antagonism, especially under severe label imbalance.**

This work proposes a multi-task learning framework that jointly models fine-grained emotional abuse behaviors and binary relationship abuse in text. To address class imbalance in the multi-label emotional behavior task, the model replaces standard binary cross-entropy with Focal Loss, optimizes a weighted joint loss across tasks, and applies post-hoc, per-label dynamic threshold tuning on a validation set to maximize macro-F1.

On the Reddit relationship abuse dataset, the framework achieves a high F1 score for binary abuse detection, while on the Unhealthy Comments Corpus it reaches an F1 of approximately 0.54 for multi-label emotional behaviors, improving substantially over the near-zero baseline reported in prior work.

These results indicate that a multi-task learning approach with imbalance-aware training and dynamic calibration can better detect unhealthy relationship patterns in text and may support future applications in content moderation and abuse intervention systems.

***Index Terms*—Intimate partner violence, Emotional abuse detection, Relationship abuse, Multi-task learning, Multi-label classification, Class imbalance, Focal Loss, Transformer models**

I. Introduction

According to the World Health Organization, nearly 30% of women experience physical or sexual abuse in their lifetime, and 1 in 7 men experience some form of partner violence. On the other hand, the rise of online platforms such as social media has increased global interaction, but has also led to a surge in harmful behaviors such as hate speech and abusive language. Under many celebrity posts, inappropriate or abusive comments have become so common that they are often normalized.

Many individuals fail to recognize that seemingly normal day-to-day behavior in relationships can gradually turn into habitual abuse, whether physical, verbal, or sexual. Such patterns can significantly impact the mental health of the affected individual in the long run.

Traditionally, abusive content has been handled through manual moderation or rule-based systems. However, these approaches are inefficient and struggle to scale with the vast amount of data generated online. Moreover, abusive language is often subtle and context-dependent, making it difficult for traditional methods to detect accurately.

To address these challenges, machine learning techniques have emerged as effective tools for abuse detection. These models learn patterns from large datasets and classify text based on the likelihood of being abusive or non-abusive. By leveraging natural language processing techniques, machine learning models can analyze textual data more efficiently and capture patterns that may be difficult for humans to identify.

II. Literature Review

The rapid growth of online platforms has led to a rise in abusive and harmful language, making abuse detection an important problem in Natural Language Processing (NLP) [2]. Online abuse, like hate speech, harassment, and offensive language, can negatively impact individuals and affect their participation in online environments [2]. The great quantity of user-generated content on social media, makes manual monitoring unfeasible. Hence, such content must be effectively recognized and managed by automated systems. This has resulted in increased research efforts toward developing reliable NLP-based solutions for abuse detection.

Abusive language includes different forms like hate speech, cyberbullying, and offensive communication [3]. Detecting such content is challenging because it depends on context, user intent, and variations in language [3]. A common classification is between explicit and implicit abuse. Explicit abuse includes direct offensive expressions, while implicit abuse is conveyed through indirect language such as sarcasm or subtle remarks [2]. Identifying implicit abuse is more difficult due to the lack of clear indicators. Different datasets capture different aspects of this problem. For example, the UCC dataset focuses on emotional categories of abuse, while Reddit-based datasets capture more direct forms of relationship abuse [8], [9].

Various approaches have been proposed for detecting abusive language. Early approaches applied rule-based systems and lexicons of offensive words, but these methods were limited in handling context [3]. Machine-learning models like Support Vector Machines and Logistic Regression enhanced their performance by leveraging features such as n-grams and TF-IDF [3]. Newer methods rely on deep learning models such as CNNs, RNNs, and transformer-based architectures that better capture the context [2]. Models like the multi-task CNN suggested in [7] have demonstrated enhanced performance by learning shared representations, and similarly, combining abuse detection with emotion classification, covered in [5], has improved the accuracy of detection.

Multi-Task Learning (MTL) is used to learn multiple related tasks together to enhance performance. MTL allows cross-task usage of a shared representation rather than building separate models [4]. This is helpful when tasks have some relation, for example, emotion detection and abuse classification. Pairing tasks increases performance and generalization, as shown in earlier work. Work in [6] examines ways in which emotional aspects may improve the quality of the detection of abusive text. In [1], similar frameworks use a multi-task approach to assess detection of both emotional abuse and relationship abuse, stressing the importance of shared learning.

The detection of abusive language also depends on the emotion involved, particularly when abuse is not directly expressed. Sarcasm, hostility, and tone are all very effective cues for detection [5], [6]. The messages we receive are impacted by politeness and emotional context when interpreting them. [4] shows that combining emotion and politeness detection increases the utility of conversational data and improves knowledge acquisition within this context. Adding such features helps to understand subtle forms of abuse that cannot be picked up through surface-level knowledge.

This still remains a work in progress, but not without some obstacles. One key challenge has been the difficulty of identifying implicit abuse, such as sarcasm or indirect expressions. This calls for a deeper contextualization [2]. Differences in datasets and annotation methods also introduce inconsistencies that affect model performance [3]. Most models do well on explicit abuse but have difficulty with subtle cases. Moreover, separate treatment of tasks limits the ability of models to capture the relationships between emotional and abusive signals. Thus, this paper suggests that better techniques are warranted and should be used, such as multi-task learning frameworks, as in [1].

III. Methodology

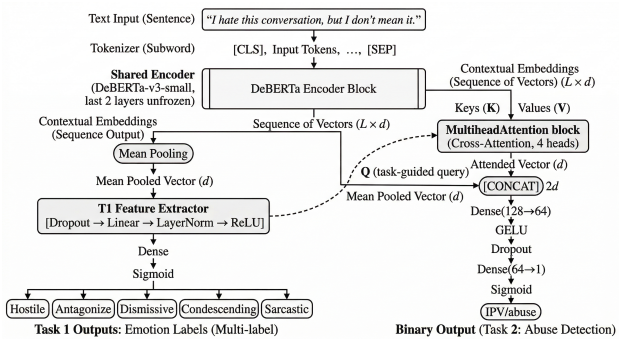

Fig. 1. Proposed multi-task learning architecture

A. Problem Formulation

In this study, the formulation of our problem is based on the framework of multi-task learning (MTL), which seeks to develop a joint model to capture fine-grained emotional toxicity and binary abusive behavior detection. From the input textual sequence x, the output generated by the model consists of a multi-label vector representing emotional labels and a binary label representing abusive behavior.

Our goal is to train a combined mapping function as follows:

$$f_\theta(x) \rightarrow \{\hat{y}^{(1)}, \hat{y}^{(2)}\} \tag{1}$$

where $\hat{y}^{(1)} \in \{0,1\}^5$ is the probabilistic representation of five emotional labels, while $\hat{y}^{(2)} \in \{0,1\}$ refers to the classification of abusive behavior. We assume that with such co-training, the fine-grained emotional semantics can serve as prior knowledge.

B. Input Sequence Representation

Each input sequence is tokenized and padded to a maximum length of 128 tokens to balance computational efficiency and contextual coverage. The sequence is represented using token IDs and an attention mask.

$$\mathbf{H} = \mathcal{E}(\mathbf{I}, \mathbf{M}) \in \mathbb{R}^{L \times d} \tag{2}$$

To obtain a fixed-size representation, masked mean pooling is used:

$$\mathbf{h}_{mean} = \frac{\sum_{i=1}^{L} M_i \cdot \mathbf{H}_i}{\sum_{i=1}^{L} M_i} \tag{3}$$

C. Task 1: Emotional Feature Learning

$$\mathbf{z}_1 = \phi(\mathbf{h}_{mean}) \tag{4}$$

$$\hat{y}^{(1)} = \sigma(\mathbf{W}_1 \mathbf{z}_1 + \mathbf{b}_1) \tag{5}$$

D. Task-Guided Cross-Attention

We use Task 1 representations to guide Task 2 through cross-attention:

$$\mathbf{Q} = \mathbf{z}_1, \quad \mathbf{K} = \mathbf{H}, \quad \mathbf{V} = \mathbf{H} \tag{6}$$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \tag{7}$$

E. Loss Functions

Task 1 (Focal Loss):

$$\mathcal{L}_1 = -\alpha(1 - p_t)^\gamma \log(p_t) \tag{8}$$

Task 2 (Binary Cross-Entropy):

$$\mathcal{L}_2 = -[y \log(\hat{y}^{(2)}) + (1 - y) \log(1 - \hat{y}^{(2)})] \tag{9}$$

F. Uncertainty-based Loss Weighting

$$\mathcal{L} = \frac{1}{2}e^{-\sigma_1}\mathcal{L}_1 + \frac{1}{2}\sigma_1 + \frac{1}{2}e^{-\sigma_2}\mathcal{L}_2 + \frac{1}{2}\sigma_2 \tag{10}$$

G. Training and Evaluation

Mini-batch stochastic gradient descent is used with gradient accumulation to stabilize updates. Threshold tuning on validation data is applied for Task 1 predictions.

IV. Results and Discussion

TABLE I
RESULTS ON TASK-GUIDED ATTENTION FOR TASK 2 (ABUSE DETECTION)

Model Variant	F1 Score	Accuracy
Full Model (with T1 guidance)	0.9052	0.9087
Without T1 Guidance	0.8727	0.8838

The effectiveness of the proposed multitasking learning approach is validated on both tasks based on F1-score and accuracy. The proposed model is particularly effective on Task 2, as indicated by the F1-score and accuracy of 0.9052 and 0.9087, respectively, under the full model conditions. It can be seen that the model performs well in distinguishing abusive and non-abusive comments, even considering subtle linguistic cues.

In order to determine the effect of task-level attention on the performance of the multitasking model, an ablation experiment is carried out through the removal of the Task 1 representations' influence on the attention module. In this case, the representation used for cross-attention will be the mean-pooled representation, thereby eliminating any influence of the task-level information.

From the above table, it is evident that the model's performance significantly reduces once the guidance of Task 1 is eliminated. The reduction in Task 2 F1-score and accuracy is 0.0325 and 0.0249, respectively. This implies that the representations generated by Task 1 play a vital role in improving the model's performance on Task 2.

The performance discrepancy confirms the theory that subtle emotional signals serve as valuable intermediary features in detecting abuse. Although the encoder model learns semantics, no explicit model exists for emotions. With the inclusion of features from Task 1, the model is able to concentrate on tokens with both semantic and emotional importance, thus enhancing the classification accuracy.

Nevertheless, the fairly high performance score of the ablated variant (F1 $\approx$ 0.87) proves that the base encoder model forms a reliable foundation. Nevertheless, the consistent improvement of the complete model shows that task-guided attention helps form inductive bias rather than just adding complexity to the model.

In summary, the findings show that the proposed model works well since it achieves significant performance increases through structured interactions among tasks. Moreover, the ablation experiment shows that the improvements result from Task 1 feature inclusion within the attention mechanism.

V. Conclusion

In this work, we proposed a multi-task learning framework for relationship abuse detection that jointly models fine-grained emotional behaviors and binary abuse classification. The model leverages a shared transformer encoder and introduces a task-guided cross-attention mechanism, where emotional representations learned in Task 1 are used to guide the detection of abusive content in Task 2.

The results demonstrate that the proposed approach achieves strong performance, particularly in binary abuse detection, with high F1-score and accuracy. The ablation study further confirms the importance of task-level interaction, showing a noticeable drop in performance when emotional features are excluded from the attention mechanism. This highlights that emotional cues act as valuable intermediate signals that enhance the model's ability to identify subtle and context-dependent abuse.

Moreover, the results indicate that while the base encoder provides a solid foundation, the integration of task-guided attention introduces a meaningful inductive bias rather than unnecessary complexity. The model benefits from structured information sharing between tasks, leading to improved generalization and more accurate predictions.

Overall, this study demonstrates that incorporating emotional understanding into abuse detection systems can significantly improve performance. Future work can focus on enhancing the modeling of implicit and nuanced emotional signals, expanding datasets, and exploring more advanced attention mechanisms to further improve detection of subtle forms of abuse in real-world scenarios.

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